Terraform backend

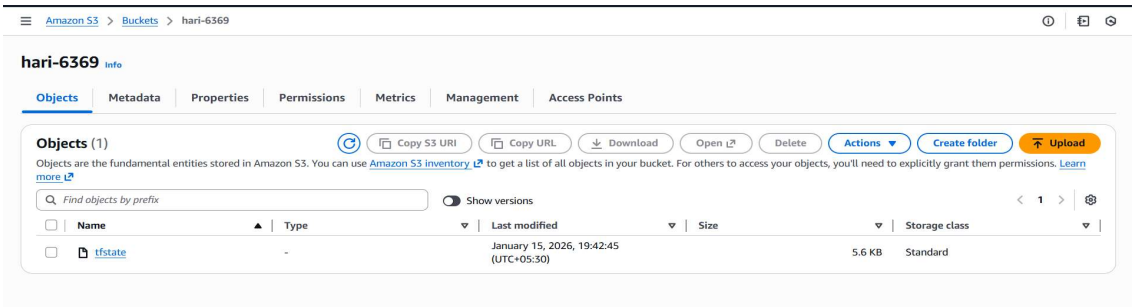
In Terraform, a backend is responsible for storing the state data files that Terraform uses to keep track of the resources it manages. By default, Terraform uses the local backend, which stores state as a local file on disk. However, you can configure a different backend to store state in a remote location, allowing multiple people to access and collaborate on the same infrastructure resources.

Terraform state:

- Maps **real cloud resources ↔ Terraform code**
- Tracks dependencies
- Prevents Terraform from recreating or deleting random stuff

Lose or corrupt state = **you lose control of your infrastructure**.

Then we have the state file created in the existed **S3 bucket**

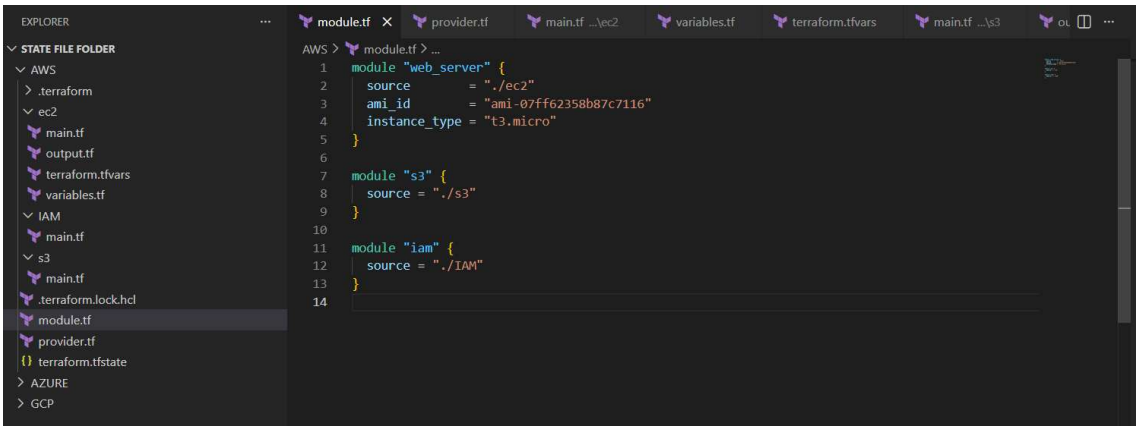


Terraform modules

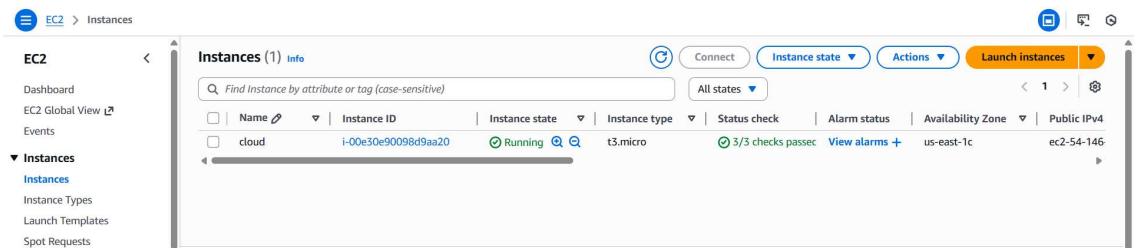
Terraform modules are reusable, self-contained packages of Terraform configuration files that simplify and standardize infrastructure provisioning. They allow you to group related resources together, making your code cleaner, more manageable, and reusable across multiple projects or environments.

I have created **module.tf** file to access the given file directory.

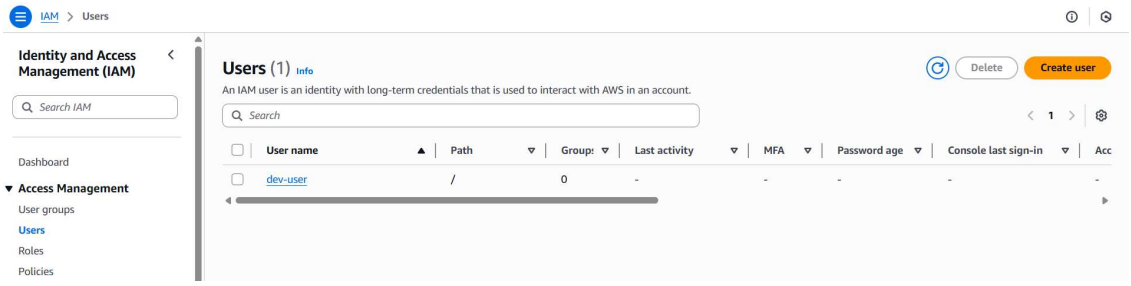
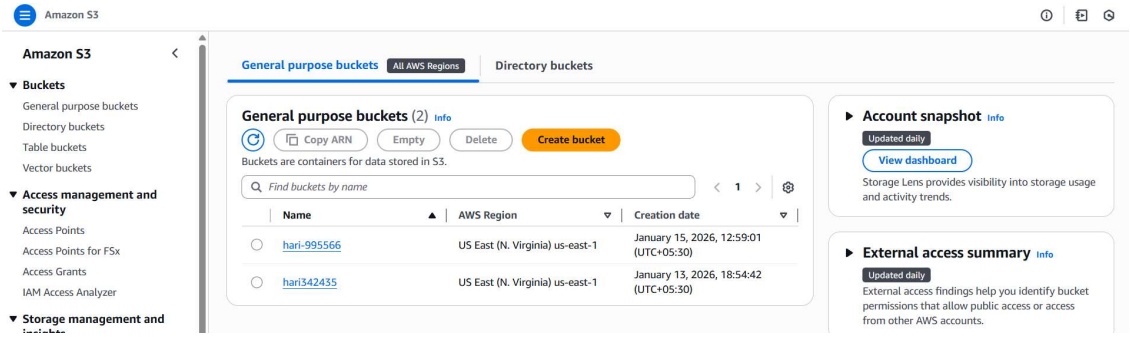
And I have created all the terraform codes for the creation of the multiple AWS services.



Created EC2 using this module.



Created IAM user.



Created S3 bucket.